

TECHNICAL REFERENCE DOCUMENT

An Open-Source 5G NR Non-Terrestrial Network Testbed

Reference Architecture, Release-17 GEO Implementation,
and Roadmap for Direct-to-Device Services

Version 1.0 · June 2026

Sabbir Ahmed

Doctoral Researcher, 6G Flagship

University of Oulu, Finland

sabbiraw.ahmed@gmail.com

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Contents

Abstract

1 Scope

2 References and abbreviations

2.1 Abbreviations

3 Background: NTN in 3GPP

3.1 The non-terrestrial challenge

3.2 Release 17 / 18 / 19 evolution

3.3 Transparent vs regenerative payloads

4 Reference architecture

4.1 System overview

4.2 Network functions

4.3 Interfaces and protocols

5 Release-17 GEO procedures

5.1 NTN timing toolkit

5.2 End-to-end call flow

6 Implementation and validation

6.1 Testbed configuration

6.2 Engineering findings

6.3 Verified results and limitations

7 Roadmap

8 Conclusion

References

Abstract

This document presents a reference architecture and a working, fully open-source testbed for 5G New Radio (NR) Non-Terrestrial Networks (NTN). It describes the system as built, an end-to-end 3GPP Release-17 geostationary (GEO) satellite attach realised entirely in software using OpenAirInterface (OAI) and Open5GS, with the satellite channel emulated in OAI's RF simulator, and sets that implementation within a broader reference framework for direct-to-device (D2D) satellite services. The testbed completes the full access and registration chain over an emulated 238.74 ms one-way GEO link: cell synchronisation, SIB19 acquisition, four-step random access, RRC connection setup, NAS registration, and PDU session establishment. The document details the NTN timing toolkit (SIB19 ephemeris, K-offset, timing advance), the network functions and interfaces, the procedures that differ from terrestrial operation, two non-obvious engineering problems encountered and resolved (including a protocol-level fix to the random-access response window), and an honest account of the emulation's limitations. A phased roadmap extends the work toward transport-mode comparison, voice feasibility, NTN-terrestrial roaming, cloud-native deployment, and low-earth-orbit (LEO) scenarios.

1 Scope

The present document specifies a reference architecture for a 5G NR Non-Terrestrial Network operating with a transparent geostationary payload, and documents its realisation as a reproducible open-source testbed. It covers the system architecture, the network functions and their roles, the interfaces and protocols, and the Release-17 procedures that enable operation across satellite-scale propagation delay.

The document is intended as both an educational reference and a record of an implemented system. Clauses 3 to 5 describe the architecture and procedures at a level applicable to any Release-17 transparent-payload NTN; clause 6 documents the specific testbed, its validation, and its limitations; clause 7 defines a roadmap of planned extensions. Where the document describes capabilities that are planned rather than implemented, this is stated explicitly.

The following are outside the scope of the present version: real radio transmission (the testbed uses RF simulation only); production-grade reliability, performance, or scale characterisation; certification or regulatory compliance; and any closed-source or commercial component.

2 References and abbreviations

This document refers to the 3GPP specification set for NTN (notably TS 38.300, TS 38.211–38.213, TS 38.331, and TS 23.501), to the relevant Release-17, 18, and 19 work, and to the open-source projects integrated in the testbed. Full citations are listed in the References clause.

2.1 Abbreviations

Term	Definition	Term	Definition
AMF	Access & Mobility Management Function	NTN	Non-Terrestrial Network
CU / DU	Central Unit / Distributed Unit	PDU	Protocol Data Unit
D2D	Direct-to-Device	PRACH	Physical Random Access Channel
DRB	Data Radio Bearer	RACH	Random Access Channel
GEO	Geostationary Earth Orbit	RAR	Random Access Response
gNB	Next-generation NodeB (5G base station)	RRC	Radio Resource Control
GNSS	Global Navigation Satellite System	SIB	System Information Block
HARQ	Hybrid Automatic Repeat Request	TA	Timing Advance
LEO	Low Earth Orbit	TN	Terrestrial Network
NAS	Non-Access Stratum	UE	User Equipment
NF	Network Function	UPF	User Plane Function

3 Background: NTN in 3GPP

3.1 The non-terrestrial challenge

A Non-Terrestrial Network places the radio access node, or the path to it, in the sky: aboard a satellite or a high-altitude platform. The defining difference from a terrestrial network is distance, and therefore delay. A geostationary satellite orbits at approximately 35,786 km altitude; the one-way propagation delay between a ground terminal and the satellite is on the order of 119–139 ms, and the full ground-to-ground path through the satellite is approximately 238.74 ms one way, or roughly 477 ms round trip. Every timing relationship in the 5G air interface, which was designed assuming a base station within tens of kilometres, must be revisited at this scale.

Two physical effects dominate. The first is the large propagation delay just described, which breaks the assumptions behind random-access timing, scheduling offsets, and feedback-based retransmission. The second is the Doppler shift caused by relative motion between the terminal and the satellite; for GEO this is negligible because the satellite is nearly stationary relative to the ground, but for LEO it is substantial. The Release-17 NTN work introduced a toolkit to compensate for both, described in clause 5.1.

3.2 Release 17 / 18 / 19 evolution

NTN support entered the 5G NR specifications in Release 17 and has matured over two subsequent releases.

Release	Theme	Key NTN capabilities
Rel-17 (2022)	Foundation	NTN introduced for NR and IoT; transparent payload; SIB19 ephemeris broadcast; K-offset scheduling; ephemeris-based timing advance; GNSS-equipped UE; L- and S-band; conditional handover for moving cells.
Rel-18 (5G-Advanced)	Enhancement	Higher bands (Ku/Ka, FR2); improved terrestrial-to-NTN handover; IoT-NTN performance and coverage; power-saving for discontinuous coverage.
Rel-19 (frozen 2025)	Architectural step	Regenerative payload (full gNB onboard the satellite); store-and-forward for delay-tolerant IoT; NB-IoT and RedCap over NTN in the 5G core; network-based location verification.

The testbed in this document implements the Release-17 transparent-payload GEO case, which is also the basis of the direct-to-device satellite services entering commercial operation today.

3.3 Transparent vs regenerative payloads

The single most important architectural distinction in NTN is the nature of the satellite payload. A *transparent* (or bent-pipe) payload acts as a radio relay: it receives the signal, shifts frequency, amplifies, and re-transmits, without decoding it. The base station remains on the ground, and the satellite simply extends its reach. A *regenerative* payload, introduced in Release 19, places signal-processing capability, part or all of the gNB, aboard the satellite itself. The testbed described here uses a transparent payload: the gNB is on the ground, and the satellite is modelled as a pure propagation delay.

4 Reference architecture

4.1 System overview

The system comprises a User Equipment, a transparent satellite relay, a ground-based gNB, and a 5G Standalone core network connected to an external data network. Figure 1 shows the end-to-end architecture as realised in the testbed, with the satellite channel emulated in software.

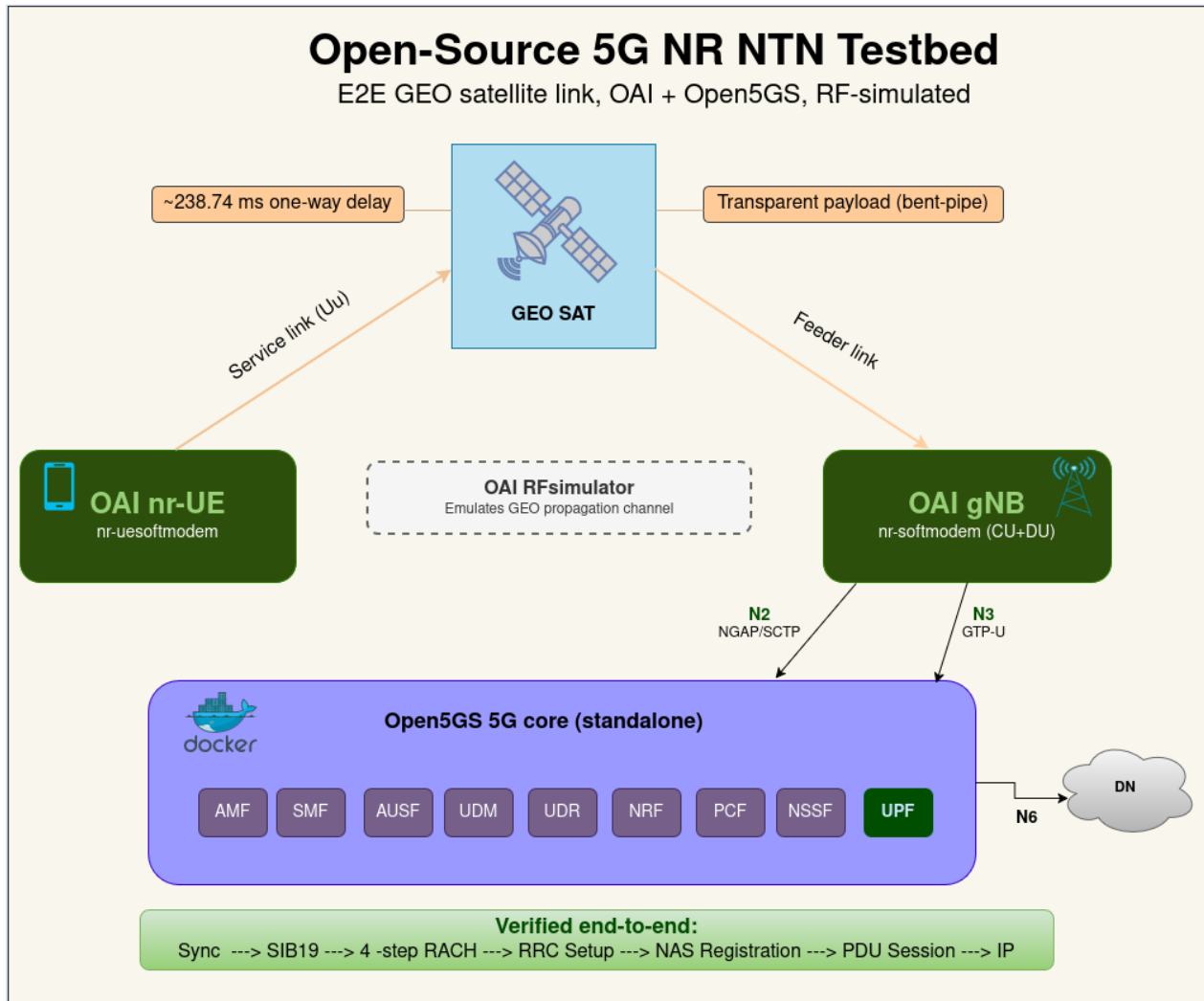


Figure 1. End-to-end 5G NR NTN testbed architecture (transparent GEO payload, RF-simulated).

Only the radio link between UE and gNB traverses the satellite. From the gNB inward, the architecture is identical to a terrestrial 5G Standalone deployment. This is the central simplifying property of the transparent-payload model: NTN-specific behaviour is concentrated in the air interface and its timing, while the core network is unchanged.

4.2 Network functions

The control plane and user plane of the 5G core are realised by the following network functions, each implemented by Open5GS in the testbed.

NF	Role
AMF	Access and mobility management; the UE's control-plane anchor. Terminates N1 (to UE) and N2 (to gNB); coordinates registration and connection management.
SMF	Session management; establishes and controls PDU sessions, assigns UE IP addresses, and programs the UPF over N4.
UPF	User-plane forwarding; carries user data between the gNB (N3) and the data network (N6). The only user-plane NF.
AUSF	Authentication server; executes the security challenge that authenticates the UE.
UDM / UDR	Subscriber data management and the underlying repository (subscriber identities and keys).
NRF	Network repository; the service directory through which NFs discover one another.
PCF	Policy control; quality-of-service and charging rules.
NSSF	Network slice selection.

4.3 Interfaces and protocols

Interface	Endpoints	Protocol	Purpose
Uu	UE ↔ gNB	NR air interface	Radio link; the only interface traversing the satellite.
N1	UE ↔ AMF	NAS	Registration, authentication, session signalling (carried over the radio).
N2	gNB ↔ AMF	NGAP / SCTP	Control-plane signalling between RAN and core.
N3	gNB ↔ UPF	GTP-U	User-plane data tunnel.
N4	SMF ↔ UPF	PFCP	Session control programming the user plane.
N6	UPF ↔ Data Network	IP	Connectivity to the external network.
SBI	Control-plane NFs	HTTP/2	Service-based interface (the NF service bus).

5 Release-17 GEO procedures

5.1 NTN timing toolkit

Release 17 introduced a set of mechanisms that allow the air interface to operate across the GEO propagation delay. The testbed exercises all of them.

Mechanism	Function
SIB19	A new system information block broadcasting the satellite ephemeris (position and velocity), the common timing advance, and the validity duration. The UE reads it to compute its timing.
K-offset (cellSpecificKoffset_r17)	A scheduling offset, in slots, that shifts the timing relationships modified for NTN so that uplink grants and responses remain consistent across the round-trip delay. The testbed uses 478 for GEO.
Common timing advance (ta-Common)	The propagation delay between the reference point and the satellite, broadcast so the UE can pre-compensate. The GEO value corresponds to 238.74 ms.
UE-specific timing advance	Computed by the GNSS-equipped UE from its own position and the ephemeris, so that its uplink transmissions arrive aligned at the gNB. For GEO this is approximately 3.67 million samples at the testbed's sample rate.
Extended RRC timers	T300, T301, T310, T311, and T319 are lengthened so connection-establishment procedures do not expire during the long round trip.
HARQ feedback disabling	A Release-17 feature permitting HARQ processes to be reused without waiting for feedback, since the round-trip delay makes feedback-based retransmission impractical at GEO scale.

5.2 End-to-end call flow

The complete attach proceeds through the following stages. Each is verified in the testbed (clause 6.3).

1. **Cell synchronisation.** The UE detects the synchronisation signal block and decodes the PBCH to acquire frame timing and the cell identity.
2. **System information acquisition.** The UE decodes SIB1, then the NTN-specific SIB19, obtaining the ephemeris, common timing advance, and K-offset. It now has the information needed to operate across the delay.
3. **Four-step random access.** The UE transmits the PRACH preamble (Msg1), applying the full NTN timing advance so the preamble arrives correctly at the gNB. The gNB responds with the Random Access Response (Msg2). The UE sends its connection request (Msg3), and the gNB resolves contention (Msg4).
4. **RRC connection setup.** The UE completes RRC setup, carrying the initial NAS message.
5. **NAS registration and authentication.** The registration request reaches the AMF over N1/N2; the AMF, AUSF, UDM, and UDR authenticate the UE; registration is accepted.
6. **PDU session establishment.** The UE requests a data session; the SMF programs the UPF over N4 and assigns an IP address; a data radio bearer is established; the UE obtains connectivity to the data network.

NOTE: Stages 1 and 2 carry the bulk of the NTN-specific behaviour. The timing advance applied at stage 3 and the response-window handling it requires are where satellite-scale delay most directly affects the protocol, and where the engineering findings of clause 6.2 arose.

6 Implementation and validation

6.1 Testbed configuration

The testbed runs on a single Ubuntu 22.04 host. The radio access network uses OpenAirInterface; the core uses Open5GS in Docker containers; the satellite channel is emulated by OAI's RF simulator configured with the GEO one-way propagation delay. All component versions are pinned by commit hash for reproducibility.

Element	Configuration
gNB	OAI nr-softmodem (CU+DU), pinned at tag 2026.w16; band 254, 5 MHz, 15 kHz SCS; SIB19 active; K-offset 478; HARQ feedback disabled.
UE	OAI nr-uesoftmodem; GNSS position configured; computes NTN timing advance from SIB19.
RF	OAI RF simulator; prop_delay = 238.74 ms on both ends.
Core	Open5GS 5G SA (Docker Compose); AMF, SMF, UPF, AUSF, UDM, UDR, NRF, PCF, NSSF; MongoDB subscriber store.

6.2 Engineering findings

Achieving a complete random-access procedure over the GEO delay required resolving two non-obvious problems. Both are documented in full in the testbed's laboratory notes.

6.2.1 RF-simulator clock divergence

The RF simulator couples the two endpoints through a lock-step sample exchange. Under default thread scheduling the two sample clocks were observed to diverge without bound, so that the gNB never received the uplink preamble. The cause was insufficient real-time scheduling priority for the simulator's sample-handling threads, consistent with a known issue in the upstream project. Running both gNB and UE under real-time scheduling (SCHED_FIFO) with the host CPU governor set to `performance` kept the clocks coupled and allowed the preamble to be received.

6.2.2 NTN-unaware random-access response window

Once the preamble was received, the procedure still failed: the UE repeatedly reported that the received RAR did not match its transmitted preamble identity. The gNB's RAR response-window check used the base `ra-ResponseWindow`, whose maximum is 80 ms, whereas a GEO round trip is approximately 477 ms. The window therefore expired before the UE could receive the RAR; the UE timed out and retried with a new preamble, and the delayed RAR for the previous preamble was discarded as a mismatch. The Msg3 scheduling path already added the NTN K-offset, but the RAR window did not. A targeted fix extends the effective response window by the cell-specific NTN K-offset, mirroring the existing Msg3 treatment; terrestrial behaviour, where the offset is zero, is unchanged. With this fix the random-access procedure completed and the full attach succeeded.

NOTE: The second finding is a protocol-level correction rather than a configuration change, and is a candidate for contribution upstream.

6.3 Verified results and limitations

The testbed completes the entire chain of clause 5.2: synchronisation, SIB19 acquisition, four-step random access, RRC setup, NAS registration, and PDU session establishment with a dedicated data radio bearer, against the live Open5GS core over the emulated GEO link. The random-access response carries the correct preamble identity, the UE registers, and a data session with an assigned IP address is established.

Two limitations of the emulation must be stated plainly:

- **RF simulation only.** No real radio is transmitted; the satellite channel is a software model of propagation delay.
- **Wall-clock latency does not reflect GEO delay.** The propagation delay is modelled correctly in the sample/timing domain, which is precisely why it governs the NTN protocol behaviour and why the timing mechanisms of clause 5.1 are exercised. However, the RF simulator is not a real-time system: it processes samples as fast as the host permits, so end-to-end packet latency measured at the IP layer reflects host processing speed rather than the physical 477 ms round trip. The delay is faithful where it determines protocol correctness; it is not a wall-clock latency emulator. This is an inherent property of the RF simulator, not a configuration error.

7 Roadmap

The implemented system (clause 6) corresponds to Phases 1 and 2 of the project. The following phases are planned extensions; they are presented here as part of the reference vision and are not yet implemented.

Phase	Objective	Status
1	5G core in containers (Open5GS)	Complete
2	gNB and UE; GEO NTN end-to-end attach	Complete
3	Reference architecture document (this document)	Complete
4	SMS over NAS (SMSoNAS) over NTN	Planned
5	Direct-to-device transport comparison: IP, Non-IP, SMSoNAS	Planned
6	Voice-over-NTN feasibility against the delay budget	Planned
7	NTN-terrestrial roaming architecture	Planned
8	Cloud-native deployment (K3s)	Planned
9–10	LEO scenario primer; AI-RAN integration sketch	Planned (optional)

A LEO testbed, which introduces time-varying delay and significant Doppler and therefore exercises a different part of the Release-17 toolkit, is planned as a separate repository.

8 Conclusion

This document has presented a reference architecture and a working open-source realisation of a 5G NR Non-Terrestrial Network with a transparent geostationary payload.

The testbed demonstrates that a complete Release-17 NTN attach, through to an established data session, can be reproduced end to end using only open-source software, and it documents the timing mechanisms, the network architecture, and the specific engineering problems that satellite-scale delay imposes on the random-access procedure. The work is offered as a reproducible educational and research artefact, with its limitations stated honestly, and as the foundation for a roadmap toward the transport modes, roaming, and orbital scenarios that define commercial direct-to-device satellite service.

References

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